

Derek Thomas

AI SOLUTIONS ARCHITECT · ML SUCCESS ENGINEER

Abu Dhabi, UAE

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Profile Summary

AI Solutions Architect with 10+ years driving production AI adoption. Expert in Generative AI, LLM Deployment & Inference, Technical Communication, and Applied Engineering. I excel in roles that blend research, engineering, and customer enablement to deliver measurable business outcomes.

Experience

AI71 (Agentic AI Platform)

Abu Dhabi, UAE

PRINCIPAL SOLUTIONS ARCHITECT

Aug. 2025 – Present

- Thought partner to developers on system design, bridging deployment realities with engineering decisions; backfilled MLOps capacity during team scaling
- Guided engineering teams on GenAI best practices: model selection in a rapidly changing landscape, token reduction, and designing for the KV-cache
- Recognized in multiple town-halls by CXO team for proactivity and raising the bar

Development

- Improved OCR performance by 10% while increasing throughput by 12x
- Reduced multi-lingual filtering error rate by 50% through dataset augmentation and fine-tuning

Vendor Management

- Led multi-lingual OCR investigation: narrowed landscape to 2 providers, developed evaluation benchmark, and selected winner against multi-faceted requirements
- Led VectorDB benchmarking under unique constraints: on-prem deployment, 5B+ documents, low RAM
- Led WebSearch provider investigation; identified and flagged potential fraud by an auto-train vendor

Customers

- Navigated on-prem deployments to regulated government entities: communicated platform requirements and negotiated configuration boundaries
- Developed platform hardware calculator to accelerate scoping and estimation for new customers

Hugging Face (World-leading Open-source AI Unicorn)

Abu Dhabi, UAE

MACHINE LEARNING SUCCESS ENGINEER

Dec. 2022 – Mar. 2025

- Drove Hugging Face's largest-ever quarter (Q1 2024): single-handedly supported Account Executives to achieve 120% of the team goal
- Advised teams on cost-efficient Foundational Model strategies: Inference Optimization, Quantization, and Compact Model adoption
- Publications & Research: authored technical blogs on Arabic RAG, Speech-to-Speech, Multi-LoRA Inference, and more

GenAI Customer Guidance

- Guided teams (startups to Fortune 500): Agentic Workflows, prompt engineering, synthetic data generation, and LLM evaluation for Finance & Healthcare
- Fine-tuned LLMs for document summarization, code-gen, automated reporting, and domain-specific Agentic RAG (PyTorch/Hugging Face)
- Deep experience profiling large-scale inference

Platform Products

- HUGS — production-ready, optimized zero-config inference microservices to accelerate deployment
- SafeCoder — self-hosted, compliant code-gen solution fine-tuned on your private code

Leadership & Communication

- Thought Leadership: high-impact speaker at GITEX Dubai and major events in Seoul and Singapore
- Cross-Functional Communication: bridged technical and non-technical discussions with C-suite, engineering, and research teams
- Team Growth & Leadership: spearheaded recruitment, growing the ML Success team by 300%

DataRobot (AutoML Platform)

Abu Dhabi, UAE

CUSTOMER FACING DATA SCIENTIST

Nov. 2021 – Sep. 2022

- First Data Scientist in the Middle East; tailored Customer Success strategies to local needs distinct from established US & Europe frameworks
- **DataRobot-Hugging Face Compatibility:** conducted largest-ever HF partner webinar in DataRobot history
- Pioneered AutoML and MLOps monitoring with Transformers (text drift, concept drift)
- Led DataRobot's first client enablement session with Transformers
- **End-to-End AI Solutions:** delivered 20+ use cases from framing to production across 10 clients
- Developed specialized training and provided on-going Data Science up-skilling
- Empathized with customers to build champions at multiple levels
- Banking: Propensity models & Next Best Offer boosted conversion rate 5x–13x per product
- Manufacturing: Machine efficiency & Material Quality drove \$1.6M in savings/yr
- Green AI: reduced CO₂ emissions nearly 200,000 tons per year
- Utilities: Solar Generation, Materials Demand Forecasting, Payment Default, et al.

UAE Government Contractor

LEAD DATA SCIENTIST / LEAD DATA ENGINEER

- Built hybrid cloud Big Data platform processing 6M documents/day as team lead
- Replaced legacy NLP/Geoparsing vendor solutions with Transformers to save \$1.3M
- Improved 20% recall (sentiment); 23% accuracy (NER); averaged across languages
- Managed data vendor relations and created an engineering solution to save \$600,000

Abu Dhabi, UAE
Feb. 2021 – Sep. 2021

G42 (AI Startup) → Unicorn)

SENIOR DATA SCIENTIST

- **Machine Translation** Arabic ↔ English: performance competitive with Google (± 3 BLEU points depending on domain)
- Led Data Engineering/Cleaning responsible for 78% of BLEU improvement from initial model
- Created fast, parallel data cleaning pipeline with Apache Airflow: 24B sentences
- **News Information Extraction:** dynamic knowledge graph inter-connecting streaming news data with 40K articles/day
- Implemented and improved SotA Entity Linking models to be compatible with BERT
- **Mixed Language OCR:** developed adaptive localized binarization filter in PyTorch; improved bounding box recognition and decreased character error rate by 50%

Abu Dhabi, UAE
Apr. 2019 – Feb. 2021

Saal.ai (AI Startup)

DATA SCIENTIST

- **Oil Price Prediction** (7-day forecasting): cleaned and feature engineered diverse time-series data of varying frequencies
- Developed NLP/AI microservices (Dockerized REST): Stance Detection (BERT semantic distance) and Graph Attention Networks for Text Classification

Abu Dhabi, UAE
Mar. 2018 – Apr. 2019

Rockwell Collins (Government Avionics)

DATA ANALYST; SYSTEMS ENGINEER; FPGA ENGINEER

- Developed a real-time data model yielding over \$15,000 in savings its first week of deployment

Cedar Rapids, Iowa, USA
Jan. 2014 – Dec. 2017

Publications

- 2020 **CIKM**, Relation Extraction with Self-determined Graph Convolutional Networks
- 2020 **ACL**, Autoencoding Keyword Correlation Graph for Document Clustering
- 2020 **SIGIR**, Attending to Inter-sentential Features in Neural Text Classification

Education

University of Louisville

MENG ELECTRICAL & COMPUTER ENGINEERING — *Magna Cum Laude* + Honours, GPA **4.0**

Louisville, KY, USA
Dec. 2011 – Dec. 2012

University of Louisville

BSC ELECTRICAL & COMPUTER ENGINEERING, MATHEMATICS MINOR — GPA **3.6**

Louisville, KY, USA
Aug. 2007 – Dec. 2011